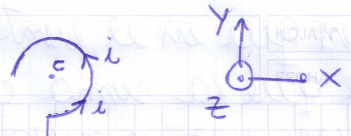


CAMPO 1: (3/4 DE ESPIRA)



BIOT SAVART $\rightarrow$

$$\vec{B} = \frac{\mu_0 \cdot i}{4\pi} \int_{-\pi/2}^{\pi} \left(\frac{R}{z} \cdot z \cdot \cos\theta, \frac{R}{z} \cdot z \cdot \sin\theta, \left(\frac{R}{z}\right)^2 \right) \frac{d\theta}{\left(\left(\frac{R}{z}\right)^2 + z^2\right)^{3/2}}$$

para $z=0 \Rightarrow \vec{B} = \frac{\mu_0 \cdot i}{4\pi} \int_{-\pi/2}^{\pi} \frac{\left(\frac{R}{z}\right)^2}{\left[\left(\frac{R}{z}\right)^2\right]^{3/2}} d\theta$; $\vec{B} = \frac{\mu_0 \cdot i}{4\pi} \cdot \frac{1}{\frac{R}{z}} \cdot \theta \Big|_{-\pi/2}^{\pi}$

$$\vec{B} = \frac{\mu_0 \cdot i}{4\pi} \cdot \frac{z}{R} \cdot \left[\pi - \left(-\frac{\pi}{2}\right) \right]; \quad \boxed{\vec{B} = \frac{3\mu_0 \cdot i}{4R} \hat{e}_z}$$

CAMPO 2: (MEDIA ESPIRA)



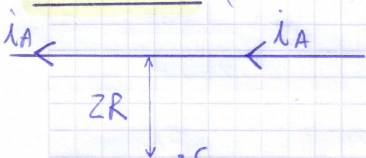
con $z=0$

$$\vec{B} = \frac{\mu_0 \cdot i}{4\pi} \left[\frac{R^2 \cdot \pi}{(R^2)^{3/2}} \right] (-\hat{e}_z) \rightarrow \text{EXPRESIÓN DEL CAMPO MAG.}$$

APLICANDO BIOT-SAVART
A UNA MEDIA ESPIRA
COMPRENDIDA ENTRE 0 y π

$$\boxed{\vec{B} = \frac{\mu_0 \cdot i}{4R} (-\hat{e}_z)}$$

CAMPO 3: (ALAMBRE RECTO E INFINITO)



$$\vec{B} = \frac{\mu_0 \cdot iA}{2\pi \cdot zR}; \quad \boxed{\vec{B} = \frac{\mu_0 \cdot iA}{4\pi R} \hat{e}_z}$$

iA : CORRIENTE DEL ALAMBRE. ELIJO UN SENTIDO DE CIRCULACIÓN

me dicen que en el punto C el CM es cero eso
implica que la suma algebraica de todos los
momentos calculados es nula.

$$0 = \text{momento 1} + \text{momento 2} + \text{momento 3}$$

$$0 = \frac{3\mu_0 i}{4R} \cdot \hat{e}_z + \frac{\mu_0 i}{4R} \cdot (-\hat{e}_z) + \frac{\mu_0 i A}{4\pi R} \hat{e}_z$$

$$0 = \frac{3\cancel{\mu_0 i}}{4\cancel{R}} (\hat{e}_z) - \frac{\cancel{\mu_0 i}}{4\cancel{R}} (\hat{e}_z) + \frac{\cancel{\mu_0 i} A}{4\cancel{\pi R}} \hat{e}_z$$

$$0 = 3i - i + \frac{iA}{\pi} ; 0 = 2i + \frac{iA}{\pi}$$

$$\boxed{A = -2i\pi}$$